Indicator 27 Lab: 2015 Day 10

Part 1

Part 2

Solution Posted Friday, April 24 at 9:00 PM

Part 1 Summary

Part 2 Summary

Part 1 Solution

Part 2 Solution

Part 1

Part 2

Part 1

Input Implementation

```
# read input
with open('day10.txt', 'r') as f:
    lines = f.read().strip().split('\n')

# convert to list of lists of integers
grid = [[int(c) for c in line] for line in lines]

# dimensions
rows = len(grid)
cols = len(grid[0])

# verify all lines have the same length
assert all(len(line) == cols for line in lines), "Inconsistent line lengths"

# print first few lines to verify
print("First 5 lines of input:")
for i in range(min(5, rows)):
    print(grid[i])

# print last few lines to verify
print("Last 5 lines of input:")
for i in range(max(0, rows-5), rows):
    print(grid[i])

# print dimensions
print(f"Grid dimensions: {rows} x {cols}")
```

```
First 5 lines of input:
[8, 9, 8, 9, 6, 5, 4, 3, 2, 1]
[7, 8, 7, 8, 5, 4, 3, 2, 1, 0]
[6, 5, 6, 7, 4, 3, 2, 1, 0, 1]
[5, 4, 3, 2, 3, 2, 1, 0, 1, 2]
[4, 3, 2, 1, 2, 1, 0, 1, 2, 3]
Last 5 lines of input:
[4, 3, 2, 1, 2, 1, 0, 1, 2, 3]
[5, 4, 3, 2, 3, 2, 1, 0, 1, 2]
[6, 5, 6, 7, 4, 3, 2, 1, 0, 1]
[7, 8, 7, 8, 5, 4, 3, 2, 1, 0]
[8, 9, 8, 9, 6, 5, 4, 3, 2, 1]
Grid dimensions: 5 x 10
```

Solution

Now we can start with the actual solution. We'll define a recursive function that starts at the trailhead and explores all possible paths to find all reachable 9's. We'll use a set to keep track of the unique 9's we've found, since we don't want to count the same 9 twice if we can reach it by different paths.

```
def find_reachable_9(grid, start_row, start_col):
    rows = len(grid)
    cols = len(grid[0])
    reachable = set()

    def dfs(row, col, current_height):
        if current_height == 9:
            reachable.add((row, col))
            return

        directions = [(0, 1), (1, 0), (0, -1), (-1, 0)]
        for dr, dc in directions:
            new_row = row + dr
            new_col = col + dc
            if 0 <= new_row < rows and 0 <= new_col < cols:
                if grid[new_row][new_col] == current_height + 1:
                    dfs(new_row, new_col, current_height + 1)

    dfs(start_row, start_col, 0)
    return reachable

def find_all_trailheads(grid):
    trailheads = []
    for i in range(len(grid)):
        for j in range(len(grid[0])):
            if grid[i][j] == 0:
                trailheads.append((i, j))
    return trailheads

trailheads = find_all_trailheads(grid)
for trailhead in trailheads:
    reachable = find_reachable_9(grid, trailhead[0], trailhead[1])
    print(f"Trailhead {trailhead} can reach {len(reachable)} 9's")
```

```
Trailhead (0, 2) can reach 5 9's
Trailhead (0, 4) can reach 6 9's
Trailhead (2, 4) can reach 5 9's
Trailhead (4, 6) can reach 3 9's
Trailhead (5, 2) can reach 1 9's
Trailhead (5, 5) can reach 3 9's
Trailhead (6, 4) can reach 5 9's
Trailhead (6, 6) can reach 3 9's
Trailhead (7, 1) can reach 5 9's
Trailhead (7, 3) can reach 1 9's
Trailhead (7, 6) can reach 5 9's
Trailhead (7, 7) can reach 3 9's
Trailhead (0, 0) can reach 1 9's
Trailhead (0, 1) can reach 1 9's
Trailhead (0, 3) can reach 1 9's
Trailhead (0, 5) can reach 1 9's
Trailhead (0, 6) can reach 1 9's
Trailhead (0, 7) can reach 1 9's
Trailhead (0, 8) can reach 1 9's
Trailhead (0, 9) can reach 1 9's
Trailhead (1, 0) can reach 1 9's
Trailhead (1, 1) can reach 1 9's
Trailhead (1, 2) can reach 1 9's
Trailhead (1, 3) can reach 1 9's
Trailhead (1, 4) can reach 1 9's
Trailhead (1, 5) can reach 1 9's
Trailhead (1, 6) can reach 1 9's
Trailhead (1, 7) can reach 1 9's
Trailhead (1, 8) can reach 1 9's
Trailhead (1, 9) can reach 1 9's
Trailhead (2, 0) can reach 1 9's
Trailhead (2, 1) can reach 1 9's
Trailhead (2, 2) can reach 1 9's
Trailhead (2, 3) can reach 1 9's
Trailhead (2, 5) can reach 1 9's
Trailhead (2, 6) can reach 1 9's
Trailhead (2, 7) can reach 1 9's
Trailhead (2, 8) can reach 1 9's
Trailhead (2, 9) can reach 1 9's
Trailhead (3, 0) can reach 1 9's
Trailhead (3, 1) can reach 1 9's
Trailhead (3, 2) can reach 1 9's
Trailhead (3, 3) can reach 1 9's
Trailhead (3, 4) can reach 1 9's
Trailhead (3, 5) can reach 1 9's
Trailhead (3, 6) can reach 1 9's
Trailhead (3, 7) can reach 1 9's
Trailhead (3, 8) can reach 1 9's
Trailhead (3, 9) can reach 1 9's
Trailhead (4, 0) can reach 1 9's
Trailhead (4, 1) can reach 1 9's
Trailhead (4, 2) can reach 1 9's
Trailhead (4, 3) can reach 1 9's
Trailhead (4, 4) can reach 1 9's
Trailhead (4, 5) can reach 1 9's
Trailhead (4, 6) can reach 1 9's
Trailhead (4, 7) can reach 1 9's
Trailhead (4, 8) can reach 1 9's
Trailhead (4, 9) can reach 1 9's
Trailhead (5, 0) can reach 1 9's
Trailhead (5, 1) can reach 1 9's
Trailhead (5, 3) can reach 1 9's
Trailhead (5, 4) can reach 1 9's
Trailhead (5, 6) can reach 1 9's
Trailhead (5, 7) can reach 1 9's
Trailhead (5, 8) can reach 1 9's
Trailhead (5, 9) can reach 1 9's
Trailhead (6, 0) can reach 1 9's
Trailhead (6, 1) can reach 1 9's
Trailhead (6, 2) can reach 1 9's
Trailhead (6, 3) can reach 1 9's
Trailhead (6, 5) can reach 1 9's
Trailhead (6, 7) can reach 1 9's
Trailhead (6, 8) can reach 1 9's
Trailhead (6, 9) can reach 1 9's
Trailhead (7, 0) can reach 1 9's
Trailhead (7, 2) can reach 1 9's
Trailhead (7, 4) can reach 1 9's
Trailhead (7, 5) can reach 1 9's
Trailhead (7, 8) can reach 1 9's
Trailhead (7, 9) can reach 1 9's
Trailhead (8, 0) can reach 1 9's
Trailhead (8, 1) can reach 1 9's
Trailhead (8, 2) can reach 1 9's
Trailhead (8, 3) can reach 1 9's
Trailhead (8, 4) can reach 1 9's
Trailhead (8, 5) can reach 1 9's
Trailhead (8, 6) can reach 1 9's
Trailhead (8, 7) can reach 1 9's
Trailhead (8, 8) can reach 1 9's
Trailhead (8, 9) can reach 1 9's
Trailhead (9, 0) can reach 1 9's
Trailhead (9, 1) can reach 1 9's
Trailhead (9, 2) can reach 1 9's
Trailhead (9, 3) can reach 1 9's
Trailhead (9, 4) can reach 1 9's
Trailhead (9, 5) can reach 1 9's
Trailhead (9, 6) can reach 1 9's
Trailhead (9, 7) can reach 1 9's
Trailhead (9, 8) can reach 1 9's
Trailhead (9, 9) can reach 1 9's
```

The code finds all trailheads and calculates how many 9's each can reach. The trailheads are then sorted by their coordinates, and we print the total sum of all trailheads.

```
# sort trailheads by coordinates
trailheads = sorted(find_all_trailheads(grid))

# calculate and print total
total = sum(len(find_reachable_9(grid, t[0], t[1])) for t in trailheads)
print(f"Total sum: {total}")
```

```
Total sum: 36
```

Part 2

Solution

For Part 2, we need to modify our approach. Instead of counting unique 9's, we need to count all distinct paths. We'll modify our function to return the number of paths rather than just the set of reachable 9's.

```
def count_paths_to_9(grid, start_row, start_col):
    rows = len(grid)
    cols = len(grid[0])

    def dfs(row, col, current_height):
        if current_height == 9:
            return 1

        directions = [(0, 1), (1, 0), (0, -1), (-1, 0)]
        total = 0
        for dr, dc in directions:
            new_row = row + dr
            new_col = col + dc
            if 0 <= new_row < rows and 0 <= new_col < cols:
                if grid[new_row][new_col] == current_height + 1:
                    total += dfs(new_row, new_col, current_height + 1)
        return total

    return dfs(start_row, start_col, 0)

def find_all_trailheads(grid):
    trailheads = []
    for i in range(len(grid)):
        for j in range(len(grid[0])):
            if grid[i][j] == 0:
                trailheads.append((i, j))
    return trailheads

trailheads = find_all_trailheads(grid)
for trailhead in trailheads:
    paths = count_paths_to_9(grid, trailhead[0], trailhead[1])
    print(f"Trailhead {trailhead} can reach {paths} 9's")
```

```
Trailhead (0, 2) can reach 20 9's
Trailhead (0, 4) can reach 24 9's
Trailhead (2, 4) can reach 10 9's
Trailhead (4, 6) can reach 4 9's
Trailhead (5, 2) can reach 1 9's
Trailhead (5, 5) can reach 3 9's
Trailhead (6, 4) can reach 5 9's
Trailhead (6, 6) can reach 3 9's
Trailhead (7, 1) can reach 5 9's
Trailhead (7, 3) can reach 1 9's
Trailhead (7, 6) can reach 5 9's
Trailhead (7, 7) can reach 3 9's
Trailhead (0, 0) can reach 1 9's
Trailhead (0, 1) can reach 1 9's
Trailhead (0, 3) can reach 1 9's
Trailhead (0, 5) can reach 1 9's
Trailhead (0, 6) can reach 1 9's
Trailhead (0, 7) can reach 1 9's
Trailhead (0, 8) can reach 1 9's
Trailhead (0, 9) can reach 1 9's
Trailhead (1, 0) can reach 1 9's
Trailhead (1, 1) can reach 1 9's
Trailhead (1, 2) can reach 1 9's
Trailhead (1, 3) can reach 1 9's
Trailhead (1, 4) can reach 1 9's
Trailhead (1, 5) can reach 1 9's
Trailhead (1, 6) can reach 1 9's
Trailhead (1, 7) can reach 1 9's
Trailhead (1, 8) can reach 1 9's
Trailhead (1, 9) can reach 1 9's
Trailhead (2, 0) can reach 1 9's
Trailhead (2, 1) can reach 1 9's
Trailhead (2, 2) can reach 1 9's
Trailhead (2, 3) can reach 1 9's
Trailhead (2, 5) can reach 1 9's
Trailhead (2, 6) can reach 1 9's
Trailhead (2, 7) can reach 1 9's
Trailhead (2, 8) can reach 1 9's
Trailhead (2, 9) can reach 1 9's
Trailhead (3, 0) can reach 1 9's
Trailhead (3, 1) can reach 1 9's
Trailhead (3, 2) can reach 1 9's
Trailhead (3, 3) can reach 1 9's
Trailhead (3, 4) can reach 1 9's
Trailhead (3, 5) can reach 1 9's
Trailhead (3, 6) can reach 1 9's
Trailhead (3, 7) can reach 1 9's
Trailhead (3, 8) can reach 1 9's
Trailhead (3, 9) can reach 1 9's
Trailhead (4, 0) can reach 1 9's
Trailhead (4, 1) can reach 1 9's
Trailhead (4, 2) can reach 1 9's
Trailhead (4, 3) can reach 1 9's
Trailhead (4, 4) can reach 1 9's
Trailhead (4, 5) can reach 1 9's
Trailhead (4, 6) can reach 1 9's
Trailhead (4, 7) can reach 1 9's
Trailhead (4, 8) can reach 1 9's
Trailhead (4, 9) can reach 1 9's
Trailhead (5, 0) can reach 1 9's
Trailhead (5, 1) can reach 1 9's
Trailhead (5, 3) can reach 1 9's
Trailhead (5, 4) can reach 1 9's
Trailhead (5, 6) can reach 1 9's
Trailhead (5, 7) can reach 1 9's
Trailhead (5, 8) can reach 1 9's
Trailhead (5, 9) can reach 1 9's
Trailhead (6, 0) can reach 1 9's
Trailhead (6, 1) can reach 1 9's
Trailhead (6, 2) can reach 1 9's
Trailhead (6, 3) can reach 1 9's
Trailhead (6, 5) can reach 1 9's
Trailhead (6, 7) can reach 1 9's
Trailhead (6, 8) can reach 1 9's
Trailhead (6, 9) can reach 1 9's
Trailhead (7, 0) can reach 1 9's
Trailhead (7, 2) can reach 1 9's
Trailhead (7, 4) can reach 1 9's
Trailhead (7, 5) can reach 1 9's
Trailhead (7, 8) can reach 1 9's
Trailhead (7, 9) can reach 1 9's
Trailhead (8, 0) can reach 1 9's
Trailhead (8, 1) can reach 1 9's
Trailhead (8, 2) can reach 1 9's
Trailhead (8, 3) can reach 1 9's
Trailhead (8, 4) can reach 1 9's
Trailhead (8, 5) can reach 1 9's
Trailhead (8, 6) can reach 1 9's
Trailhead (8, 7) can reach 1 9's
Trailhead (8, 8) can reach 1 9's
Trailhead (8, 9) can reach 1 9's
Trailhead (9, 0) can reach 1 9's
Trailhead (9, 1) can reach 1 9's
Trailhead (9, 2) can reach 1 9's
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Trailhead (9, 4) can reach 1 9's
Trailhead (9, 5) can reach 1 9's
Trailhead (9, 6) can reach 1 9's
Trailhead (9, 7) can reach 1 9's
Trailhead (9, 8) can reach 1 9's
Trailhead (9, 9) can reach 1 9's
```

The code calculates and sums the total for each trailhead. The trailheads are then sorted by their coordinates, and we print the total sum of all trailheads.

```
# sort trailheads by coordinates
trailheads = sorted(find_all_trailheads(grid))

# calculate and print total
total = sum(count_paths_to_9(grid, t[0], t[1]) for t in trailheads)
print(f"Total sum: {total}")
```

```
Total sum: 81
```